

de-Broglie wave

In 1924, Luis de-Broglie proposed that *matter possesses wave as well as particle properties*. This wave is called de-Broglie wave.

A photon of light of frequency has the momentum

$$p = \frac{h\nu}{c} \quad \Leftrightarrow \quad p = \frac{h\nu}{v\lambda} \quad \therefore \quad p = \frac{h}{\lambda}$$

The wavelength of a photon is therefore specified by the relation

$$\lambda = \frac{h}{p} \quad (1)$$

This wavelength is called de-Broglie wavelength.

de-Broglie suggested that equation (1) is completely general one that applies to material particle as well as photons.

The momentum of a particle of mass m and velocity v is given by

$$p = mv$$

and its de-Broglie wavelength is accordingly

$$\lambda = \frac{h}{mv} \quad (2)$$

Thus, the greater the particle's momentum, the shorter its wavelength.

We know,

$$E = \frac{1}{2}mv^2 \quad \Leftrightarrow \quad v = \sqrt{\frac{2E}{m}}$$

Again, we know

$$\lambda = \frac{h}{mv} \quad \Leftrightarrow \quad \lambda = \frac{h}{m\sqrt{\frac{2E}{m}}} \quad \Leftrightarrow \quad \lambda = \frac{h}{\sqrt{2mE}}$$

This is the expression for the Broglie wavelength in terms of energy.

Wave velocity and group velocity

The **phase/wave velocity** of a wave is the rate at which the phase of the wave propagates in space. This is the velocity at which the phase of any one frequency component of the wave travels.

$$v_p = \frac{\omega}{k}$$

The **group velocity** of a wave is the velocity with which the overall shape of the waves' amplitudes-known as the modulation or envelope of the wave-propagates through space.

The group velocity v_g is defined by the equation:

$$v_g = \frac{d\omega}{dk}$$

Where, ω is the wave's angular frequency (usually expressed in radians per second), and k is the angular wave number (usually expressed in radians per meter).

Mathematical Problems

Problem-1: If the de-Broglie wavelength of an electron is equal to $9 \times 10^{-10} \text{ m}$. Calculate its kinetic energy.

Solution: We know

$$\lambda = \frac{h}{\sqrt{2mE}} \quad \Leftrightarrow \quad E = \frac{h^2}{2m\lambda^2}$$

$$E = \frac{(6.63 \times 10^{-34})^2}{2 \times 9.1 \times 10^{-31} \times (9 \times 10^{-10})^2}$$

$$E = 2.98 \times 10^{-19} \text{ J}$$

$$E = \frac{2.98 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV} = 1.8625 \text{ eV (Ans)}$$

Here,

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$\lambda = 9 \times 10^{-10} \text{ m}$$

$$E = ?$$

Problem-2: If the de-Broglie wavelength of Proton is equal to $6000 \text{ \AA}$. Calculate the energy of Proton.

Problem-2: If the velocity of proton is $\frac{1}{20}$ th of the velocity of light, then calculate the de-Broglie wavelength.

Compton Effect

When an energetic photon strikes an electron (assumed to be initially at rest), it is scattered away from its original direction of motion while the electron receives an impulse and begins to move. In the collision, the photon may be regarded as having lost an amount of energy that is the same as the kinetic energy gained by the electron. Consequently, the scattered photon has less energy than the incident photon. Therefore, the frequency of the scattered photon is shorter than that of incident photon. This alteration in frequency or wavelength of the photon due to the collision with electron is called Compton Effect.

Compton Equation

Let an energetic (X-ray) photon of energy $E = h\nu$ and momentum $h\nu/c$ strikes a free electron at rest. The energy of incident photon will be divided into energy of scattered photon $= h\nu'$ and energy of electron $= K.E.$

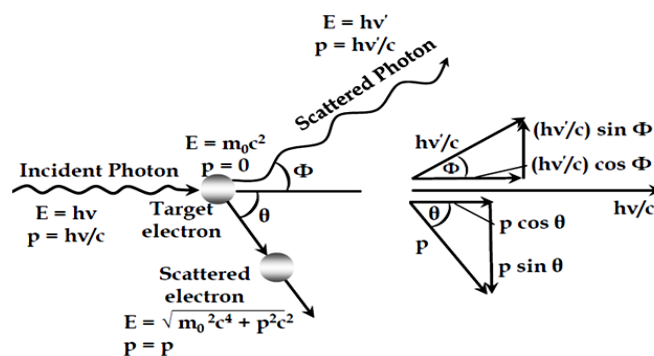


Fig. 1. Compton Effect.

$$\lambda' - \lambda = \frac{h}{m_0c} (1 - \cos \varphi)$$

This Equation is the expression for the change in wavelength when an incident radiation of wavelength is scattered by a free electron by an angle φ .

The quantity $\frac{h}{m_0c} = 0.024 \text{ \AA} = \lambda_c$ is called Compton wavelength of scattering particle.

Finally,
$$\lambda' - \lambda = \lambda_c (1 - \cos \varphi)$$

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